Ellen Su

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Education

New York University *2024 - present*

- Ph.D. Student, Center for Data Science
- Advisor: Todd Gureckis

Princeton University *2019 - 2023*

- B.S. Computer Science
- Advisor: Tom Griffiths
- Thesis: *Revealing the Priors of Deep Learning Models Through Iterated Learning*

Publications

Su, E., Srinivasan, D., Ho, M.K., & Gureckis, T.M. (2026). Language Shapes Experiential Learning in Humans and Machines. *Preprint.* https://osf.io/preprints/psyarxiv/8ahkm_v1.

Su, E.*, Potapczynski, A.*, Qiu, S., Hughes, E. & Gordon Wilson, A. (2026). Epiplexity Guided Data Selection and Generation for Out-of-Distribution Generalization. *NeurIPS 2026.*

Srinivasan, D., **Su, E.**, Ho, M.K., & Gureckis, T.M. (2026). Timing of Guidance Shapes Experiential Learning. *9th Annual Conference on Cognitive Computational Neuroscience*.

Su, E.*, Legris, S.*, Gureckis, T.M., & Ren, M. (2025). Opinion: Learning Intuitive Physics Requires More than Visual Data. *NeurIPS Embodied World Models for Decision Making Workshop 2025.*

Su, E., Ho, M.K., & Gureckis, T.M. (2025). Integration of Language and Experience via the Instructed Bandit Task. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society.*

Su, E., Vellore, A., Chang, A., Mura, R., Nelson, B., Kassianik, P., & Karbasi, A. (2024). Extracting Memorized Training Data via Decomposition. https://arxiv.org/abs/2409.12367.

Arevalo, J.*, **Su, E.***, Carpenter, A., & Singh, S. (2024). MOTIVE: A Drug-Target Interaction Graph For Inductive Link Prediction. *NeurIPS Datasets and Benchmarks 2024 (spotlight).*

Arevalo, J., **Su, E.**, Ewald, J.D., Van Dijk, R., Carpenter, A., & Singh, S. (2024). Evaluating batch correction methods for image-based cell profiling. *Nature Communications, 15.*

** equal contribution*

Awards / Invited Talks

2026	Meta AI Mentorship (AIM) Program Fellowship
2026	Brains, Minds, & Machines NIH Scholarship
2025	Computation and Decision-Making Lab, NYU
2025	Human and Machine Learning Lab, NYU
2024	Spotlight, NeurIPS 2024 (top 2.08% of submissions)
	MOTIVE: A Drug-Target Interaction Graph For Inductive Link Prediction.
2023	Princeton CS Outstanding Student Teaching Award

2022 Princeton CS Grace Hopper Celebration Grant
2021 Princeton Summer Research Fellowship

Research

Meta Fundamental AI Research (FAIR) New York, NY
Visiting Researcher, Mentors: Florian Bordes and Adina Williams *Sept 2026 - Present*
- Advancing physical reasoning in video models

Robust Intelligence San Francisco, CA
Research Intern, PI: Amin Karbasi *Summer 2024*
- Developed a decomposition jailbreak to extract training data from production LLMs

Broad Institute of MIT and Harvard Cambridge, MA
ML Research Associate, PIs: Anne Carpenter, Shantanu Singh *2023 - 2024*
- Used graph neural networks to advance drug-target interaction (DTI) discovery
- Produced a co-first author spotlight publication in NeurIPS and second author publication in Nature Communications

Princeton University Princeton, NJ
Senior Thesis, PI: Tom Griffiths *2022 - 2023*
- Used convolutional neural networks (CNNs) in iterated learning chains to identify inductive biases
- Demonstrated that machine priors manifested in sequential decision-making behavior
- Presented at the Program in Applied and Computational Mathematics Symposium

Junior Paper, PI: Ben Raphael *2021 - 2022*
- Developed to a machine learning algorithm that predicts copy number variation mutations from spatial transcriptomics data

Teaching

DS-GA1012: LLMs: Evaluations and Applications NYU, NY
Graduate Course TA & Guest Lecturer *Spring 2026*
Princeton Tutoring Princeton, NJ
Math/Physics/CS Tutor *2021 - 2024*
COS495: Web3/Blockchains Princeton, NJ
Undergraduate Course Assistant *Spring 2023*

Service

Princeton First Aid and Rescue Squad Princeton, NJ
National Registry of Emergency Medical Technicians *2021 - 2023*

Skills and Languages

Interests	Machine learning, cognitive science, artificial intelligence
Languages (program)	Python (PyTorch, JAX), Java, JavaScript, C, SQL, CSS
Languages	English (native), Mandarin (native), Spanish (advanced)